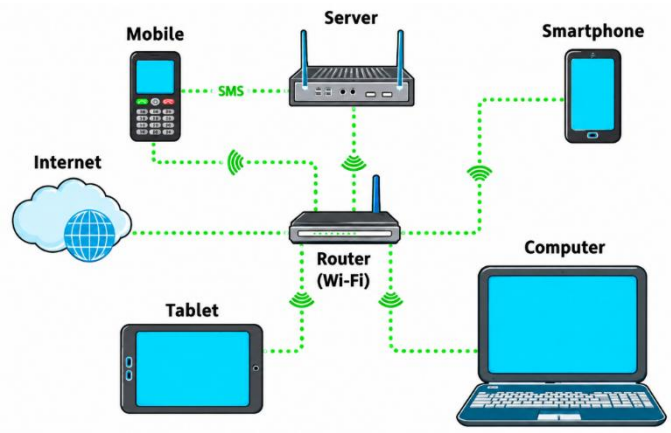
Section 1

Networking basics — LAN, WAN, routers, switches

Networking: Networking is the **process of connecting computers, devices**, or people so they can communicate and share information or resources (Data).

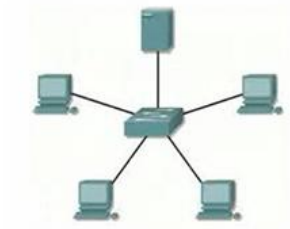
Why we use Network

- Share info or data
- Share Hardware and Software
- Emails or message
- Audio and Video calls
- Share printers
- Share Hard drives



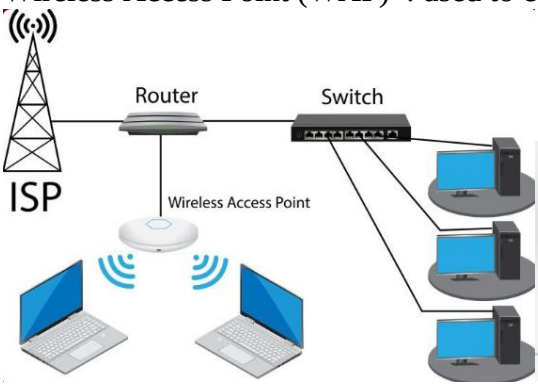
Switch: is hardware that connects devices (like computers, printers, and gaming consoles) within a Local Area Network (LAN)

Example of switch





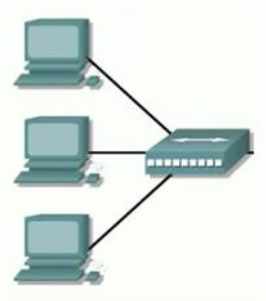
Switch talks to a different devices across your local area network. Sends your request to a relevant network device.
Wireless Access Point (WAP) : used to connect wireless devices.

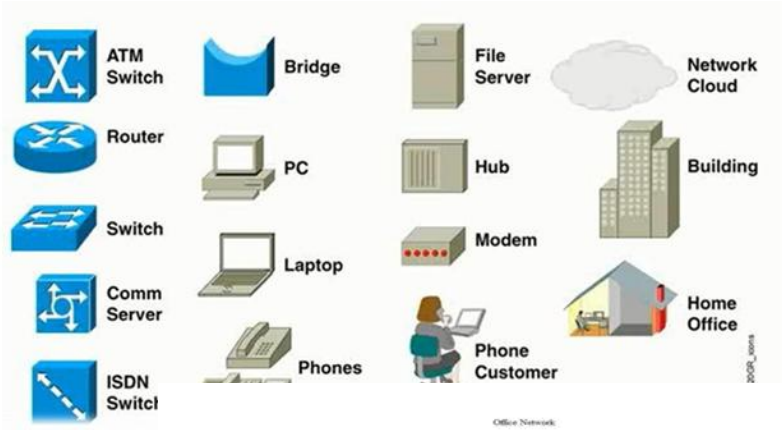




- **Hub:** Connects group of hosts

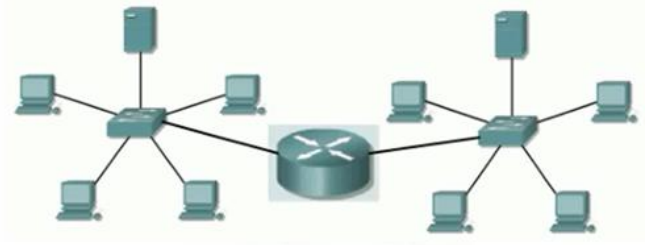
Example of Hub

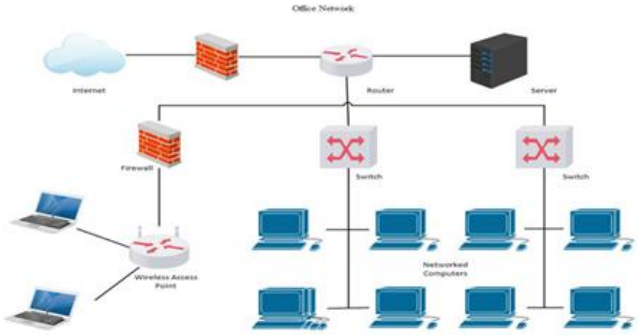




- **Router:** connects different networks or distance networks

Example of Router

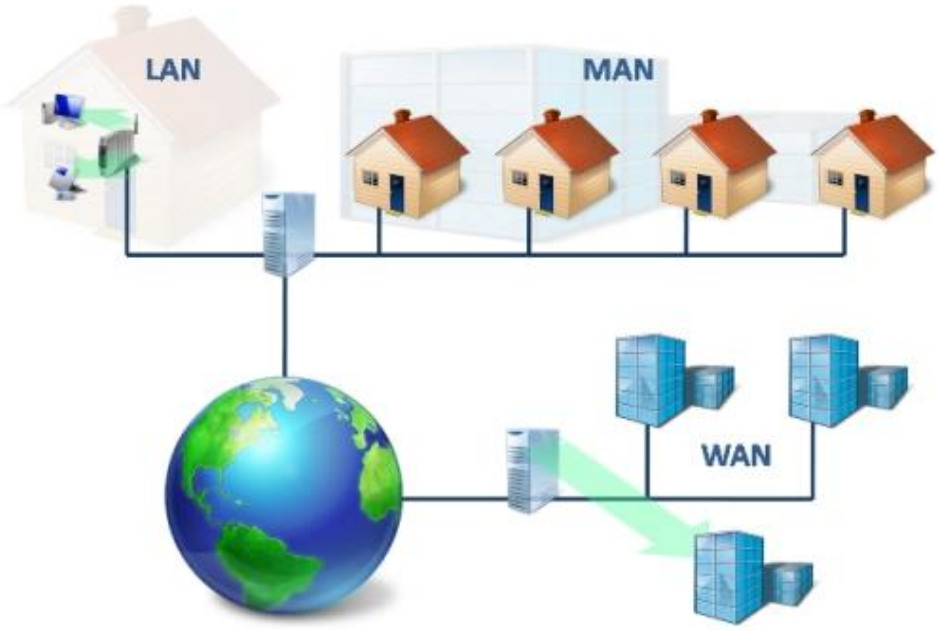




Local Area Networks (LANs) -> Operate within a limited geographic area such as office or Home

Metropolitan-area Networks (MANs) ->

Wide-area Networks (WANs) -> Operate over a large geographical area or connect LANs.



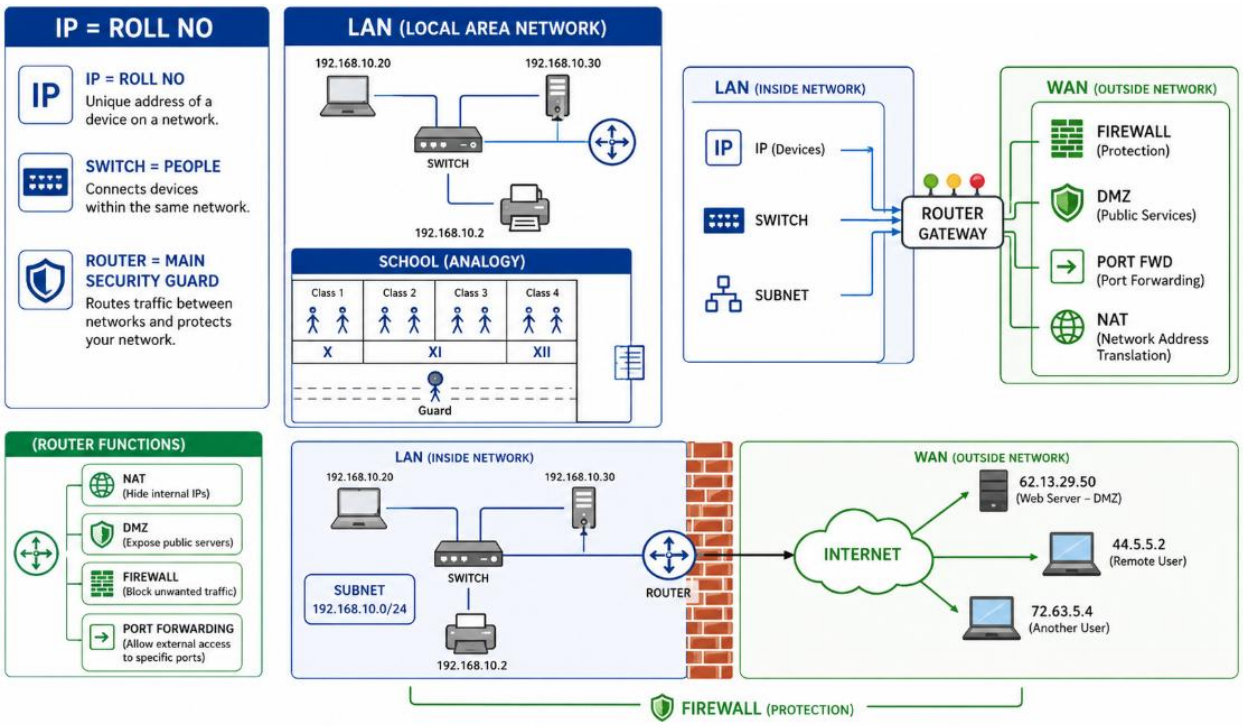
Bandwidth:

Speed of internet is called Bandwidth.

Firewall



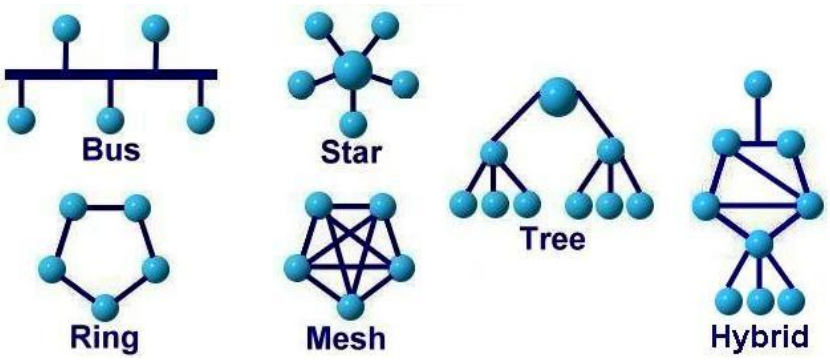
Firewall has DMZ (Demilitarized Zone ad-hock or direct connection will be reported to DMZ) and Port FWD (identifies port for our local introduced (subnet) devices).



Measuring Bandwidth

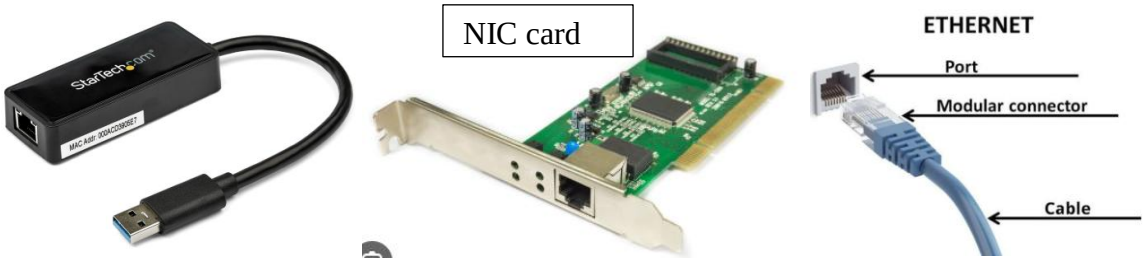
Bits per second	Abbreviation	Equivalence
Bits Per Second	bps	1 bps=fundamental unit of bandwidth
Kilobits Per Second	kbps	1kpbs=~1,000 bps=10^3 bps
Megabits Per Second	Mbps	1 Mbps=~1,000,000 bps =10^6 bps
Gigabits Per Second	Gbps	1 Gbps=~1,000,000,000 bps =10^9 bps
Terabits Per Second	Tbps	1 Tbps=~1,000,000,000,000 bps= 10^{12} bps

Topologies (Shape of Possible Networks):
Structure in network is called topology.
Mainly used are **star** and **tree**

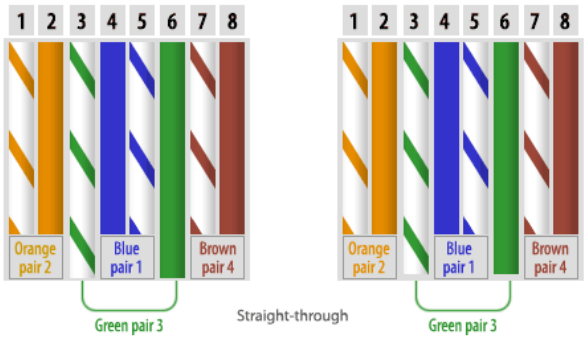


Cable Types

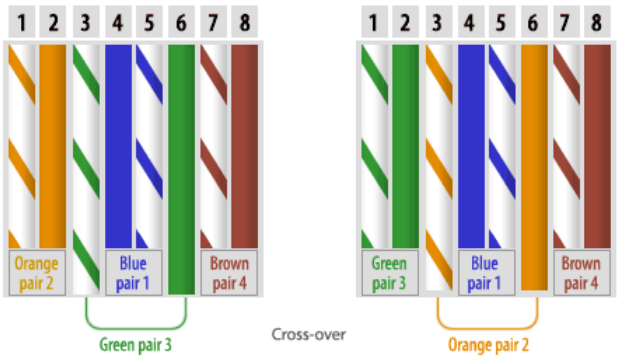
Cat 5 (Category 5) and **Cat 5e (Category 5 enhanced)** are Ethernet cables that use twisted pairs of copper



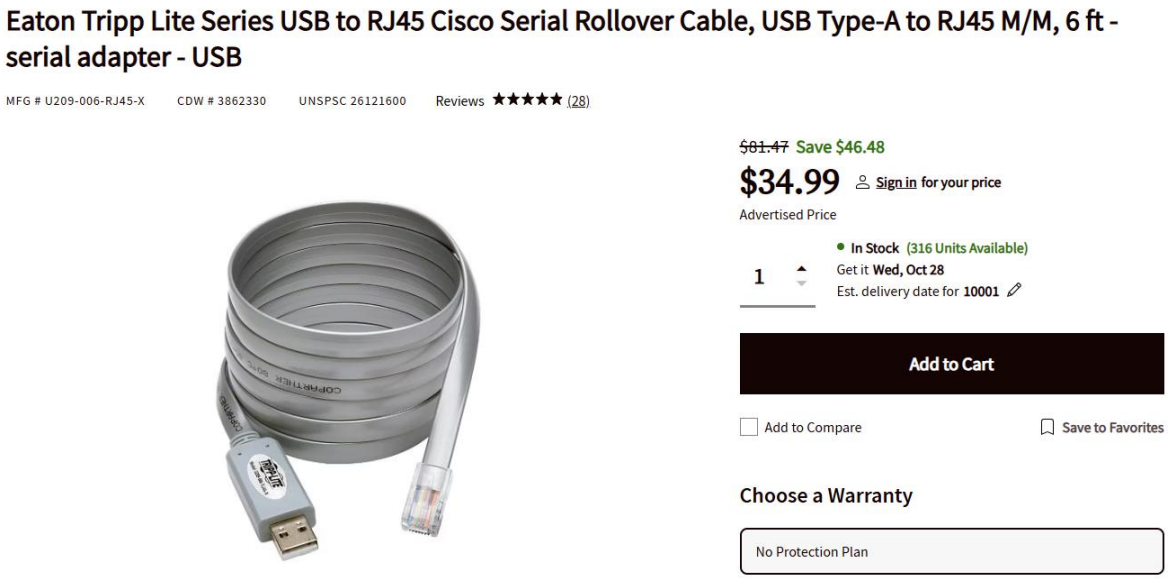
- **Straight through**: connects two different type of devices such as computer to switch, Router to switch. A straight-through cable has the same wiring standard on both ends**.** (**RJ45** connector)



- **Cross over** : connects two like network devices



Rollover cable (Console cable)



Generally this cable is used for the configuration of router and switch. Which dosent need ip address

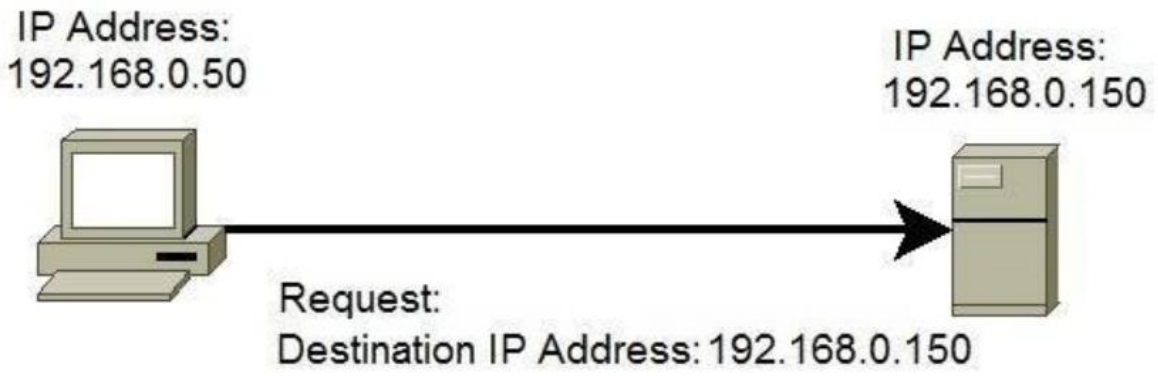
Protocols- of transport layer

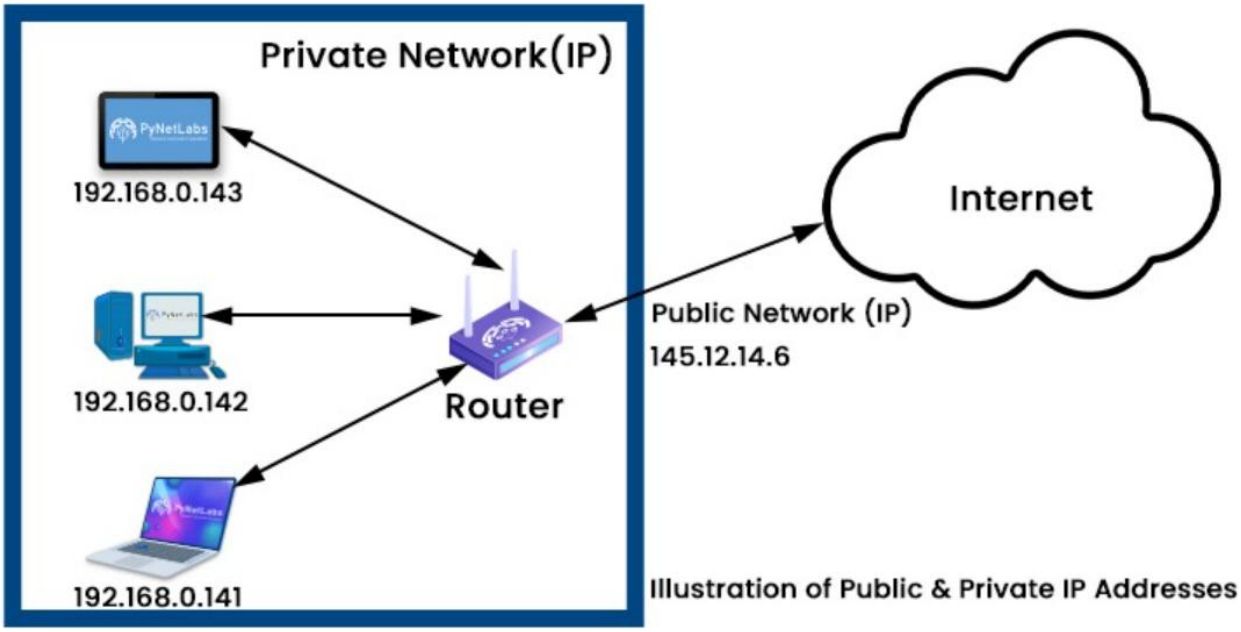
- At the **Transport Layer**; you can find these common protocols:
- **SPX** – Sequenced Packet Exchange
- **TCP** – Transmission Control Protocol
- **UDP** – User Datagram Protocol
- **SCTP** – Stream Control Transmission Protocol

Protocols – of Network Layer

- At The Network Layer You Can Find These Common Protocols:
- ATP – Apple Talk Transaction Protocol
- IPV4 – Internet Protocol V4
- Ipv6 – Internet Protocol
- IPX – Internetwork Packet Exchange
- ICMP – Internet Control Message Protocol
- IGMP – Internet Group Management Protocol
- OSPF – Open Shortest Path First

IP addressing — IPv4, IPv6, Subneting





Any network needs 3 things IP, Subnet and Gateway (IP address of Router is called gateway)

LAN Card has two addresses as below

✓ IP Address
✓ MAC Address

IP (internet protocol) Address:

MAC (Media access control) Address

A changeable logical address assigned to acms computer or device to identify it and communicate with other devices on a network.

✓ once cabling id done, we must assign IP address for all the network computers. After that the network computers can communicate, share, apply policy, and sending message etc. to others.

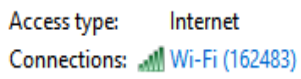
CMD = > getmac to get current device mac address (ipconfig)
The TCP/IP suite defined five blocks of addresses, called **IP address classes**, for specific network uses and sizes.

Address Class	Range	Default Subnet Mask	Networks/Nodes
Class A	1.0.0.0 to 127.255.255.255	255.0.0.0	126 networks of up to 16,777,214 nodes each
Class B	128.0.0.0 to 191.255.255.255	255.255.0.0	16,382 networks of up to 65,534 nodes each
Class C	192.0.0.0 to 223.255.255.255	255.255.255.0	2,097,150 networks of up to 254 nodes each
Class D	224.0.0.0 to 239.255.255.255	None	All members of the multicast session share the same IP address
Class E	240.0.0.0 to 255.255.255.255	None	Strictly for research and experimentation purposes

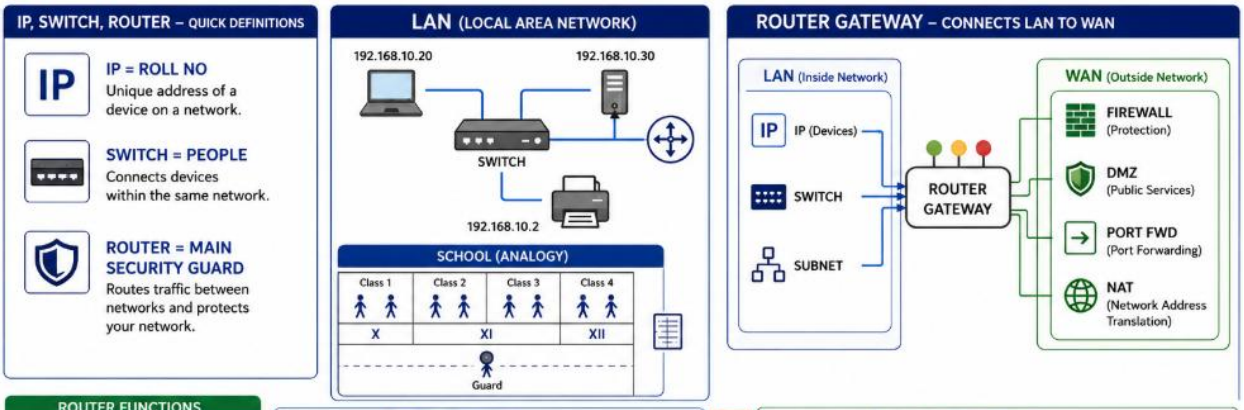
Ex: IP 192.168.1.0
Ex: Sub netmask 255.255.255.0

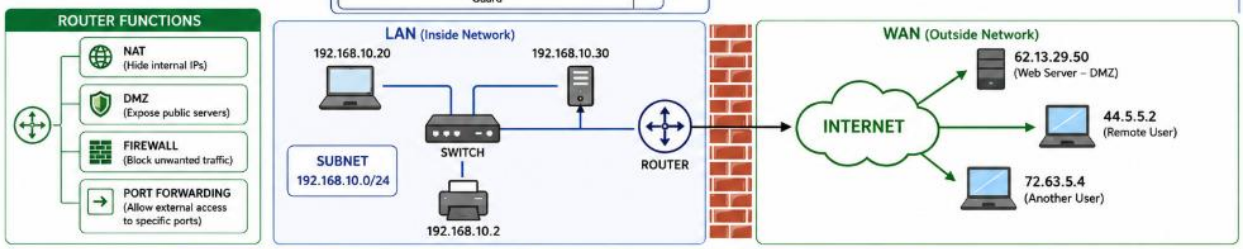
Means we have 3 octet of Networks and 1 Host. So N.N.N.H so we can't change Networks but we can change Host and also we can do sub-netting.

If you want assign new IP address:
Control Panel, Network and Internet – Network and Sharing Center- Properties – TCP/ip4

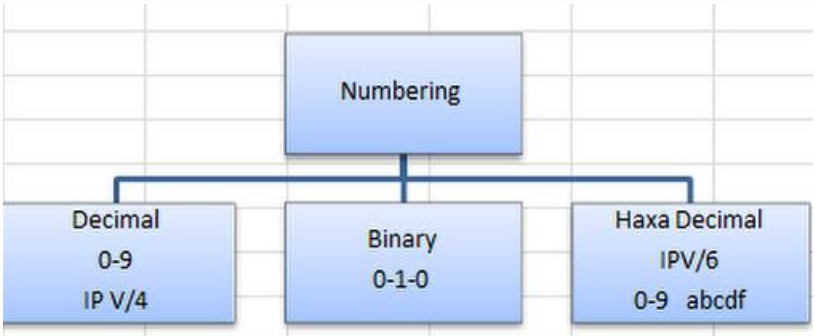


IPv4 has totally 32 – bit logical address









Subnet	1	2	4	8	16	32	64	128	256
Host	256	128	64	32	16	8	4	2	1
Subnet Mask	/24	/25	/26	/27	/28	/29	/30	/31	/32
	0	0 128 128	128 64 192						

192.168.4.0

Make 3 subnets

Total subnets 4
Total host IDs 64 - 2 = 62 including network ID and broadcast ID
Subnet Mask 255.255.255.192

Network ID	Subnet Mask	Host ID Range	# of Usable Host	Broadcast ID
192.168.4.0	/26	192.168.4.1-192.168.4.62	64- 2 = 62	192.168.4.63
0+64 = 64 192.168.4.64	/26	192.168.4.65-192.168.4.126	62	192.168.4.127
64+64 = 128 192.168.4.128	/26	192.168.4.129-192.168.4.190	62	192.168.4.191
128+64 = 192 192.168.4.192	/26	192.168.4.193-192.168.4.254	62	192.168.4.255

Subnetting	1	2	4	8	16	32	64	128	256
Hosts	256	128	64	32	16	8	4	2	1
Subnet Mask	/24	/25	/26	/27	/28	/29	/30	/31	/32

192.168.1.0

I need 8 subnetting?

32−2=30

previous number plus Hosts num

0	128	192	224
0	128	64	32
	128	192	224

255.255.255.0

255.255.255.224

Subnet	Network ID	Usable Host Range	Broadcast ID
1	192.168.1.0/27	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32/27	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64/27	192.168.1.65 – 192.168.1.94	192.168.1.95
4	192.168.1.96/27	192.168.1.97 – 192.168.1.126	192.168.1.127
5	192.168.1.128/27	192.168.1.129 – 192.168.1.158	192.168.1.159
6	192.168.1.160/27	192.168.1.161 – 192.168.1.190	192.168.1.191
7	192.168.1.192/27	192.168.1.193 – 192.168.1.222	192.168.1.223
8	192.168.1.224/27	192.168.1.225 – 192.168.1.254	192.168.1.255

CCNA-Homework:

Subneting	1	2	4	8	16	32	64	128	256
Host	256	128	64	32	16	8	4	2	1
Subnet Mask Base	/24	/25	/26	/27	/28	/29	/30	/31	/32

0

128+0 = 128

192.168.0.0/25

we need 2 subnetting
Total Host = 128
Total Usable Host = 128-126
subnet Mask
255.255.255.128

Network ID	Subnet Mask Base	Host ID Range	Broadcast ID
192.168.0.0	/25	192.168.0.1 - 192.168.0.126	192.168.0.127
192.168.0.128	/25	192.168.0.129 - 192.168.0. 254	192.168.0.255

Note: to calculate broadcast ID we subtract 1 from Host

Note: to make Host range we add total usable host to Network ID such as 0+126 or 128+126 = 254

Subnet defines your local area network Range. Subletting are private IP address that defines some one's computer.
All classes are divided into two Parts Private IPs and Public IPs

ICANN has set aside three nonroutable address ranges that a company can use internally to enable their network nodes to communicate with one another using TCP/IP.

The private, nonroutable IP address ranges are:

10.0.0.0	to	10.255.255.255
172.16.0.0	to	172.31.255.255
192.168.0.0	to	192.168.255.255

Custom subnet masks topic

Public and Private IPs

128	16	0	1	128	31	0	1

We can make 15 networks in B class

222.0.0.1 This public IP other than the range of Private IP.

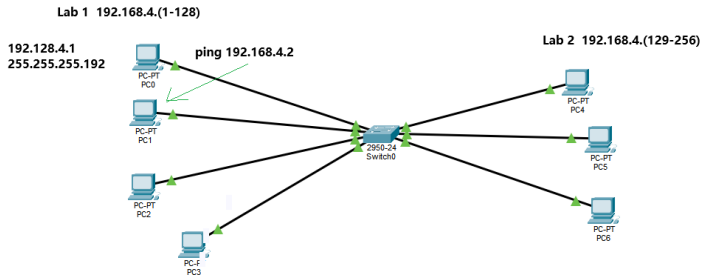
SUBNET 192.168.0.2 - 192.168.0.254

Cisco

Cisco Systems was founded in **1984** by married couple **Leonard Bosack and Sandy Lerner**, who worked at **Stanford University**. The first thing they created was gateway server to communicate. They helped develop networking technology that allowed computers on **different networks and departments to communicate**, which led to the creation of **Cisco routers**. The company was named **Cisco**, derived from **San Francisco**, California. **Cisco Packet Tracer**: is a virtual networking lab used to practice Cisco networking and CCNA skills.

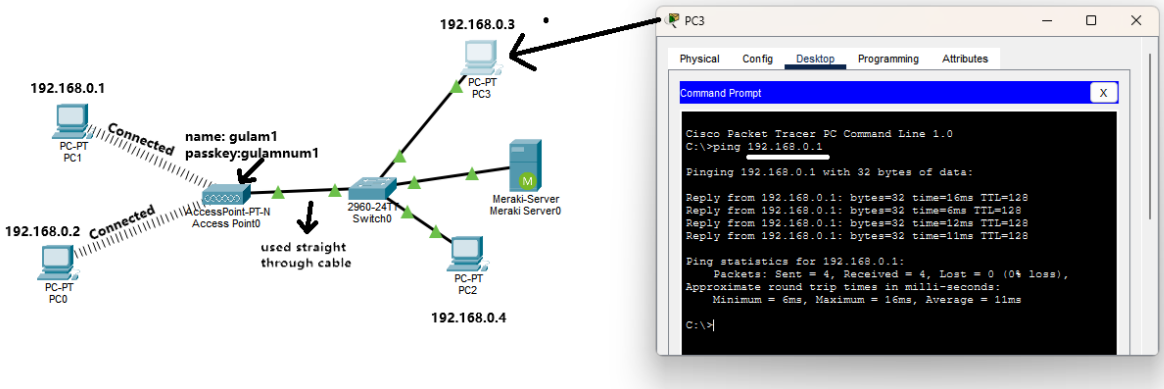
Ex: 8 bits C class IP subnetting 192.168.4.0

192 sub netting



Switch + Access Point Setup

LAN wireless + line



Reconnaissance Attacks

Discovery and mapping of systems to gather information.

Intruders look for vulnerable services.

Access Attacks

Exploit known vulnerabilities

Password Attacks

Repeated attempts, Brute-force

Disable on unsuccessful logins

OTP (One Time Password)

Strong Password



Section 2

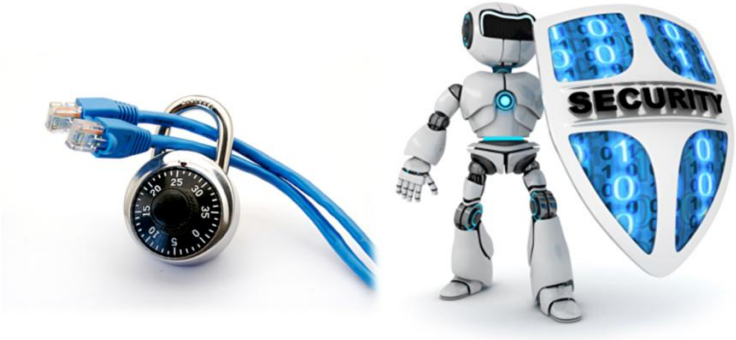
Client and Server
Server: is the computer that controls clients.
DNS Server: converts name.com to IP and IP to name.com
Active Directory
Set of rules used to create users, and rules for the clients.

Characteristics of Network

- Speed: How fast data transmitted over the network.
- Cost: General cost of components, installation, and maintenance.
- Security: How secure the network?
    - ([Number of minutes in a year-downtime]/[Number of minutes in a year])*100=Percentage availability
    - 15 minutes downtime =99.997
- Scalability and Reliability: How well your network can accommodate more users and MTBF (Mean time between failures)

Need for Network Security

- 60 to 80 percent of network misuse comes from inside
- Open and close Network
- Closed Network does not allow a connection to public networks



Classes of Attack

- Passive: Traffic analysis, monitoring communication
- Active: Attempts to circumvent or break protection features i.e. DOS
- Close-in: Attempting to go close physical proximity to networks, systems, or facilities
- Insider: Malicious insiders intentionally eavesdrop, steal, or damage information
- Distributed: Malicious modification of hardware or software during distribution with back door installed

Mitigating Common Threats



Physical Installations



✓ Room Look
✓ No access from ceiling
✓ Electronic access control (Biometric)
✓ CCTV (Closed-circuit television)
✓ Temperature and humidity control systems (HVAC) (Heating,ventilation,and air conditioning)
✓ Electrostatic and magnetic interference
✓ UPS and backup generator



Note: Lock is not the Preventive control



Network Layers OSI (Open Systems Interconnection)

7 — Application Layer
✓ Message format, Human-Machine Interfaces

6 — Presentation Layer
✓ Coding into 1s and 0s; encryption, compression

5 — Session Layer
✓ Authentication, permissions, session restoration

4 — Transport Layer
✓ End-to-end error control

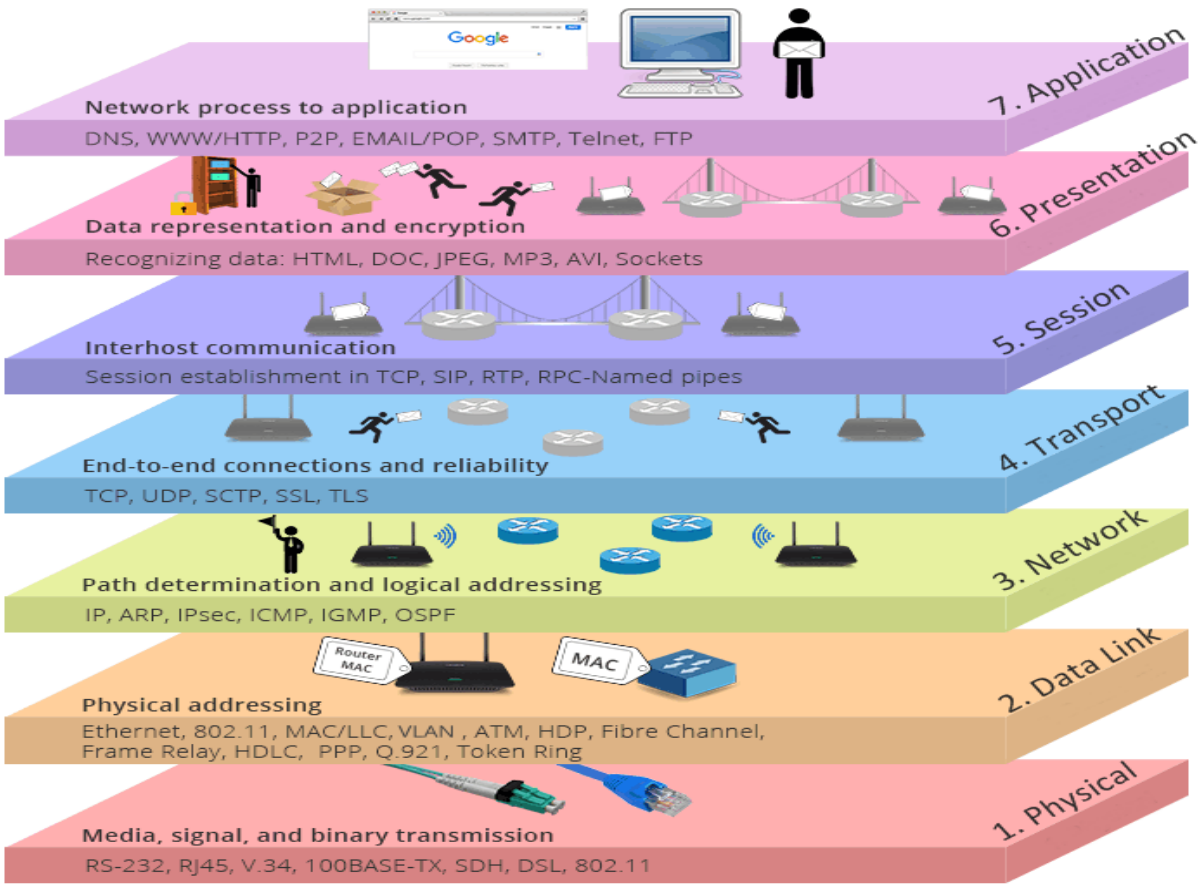
3 — Network Layer
✓ Network addressing; routing or switching

2 — Data Link Layer
✓ Error detection, flow control on physical link

1 — Physical Layer
✓ Bit stream: physical medium, method of representing bits

UPPER LAYERS: 7, 6, 5
TRANSPORT SERVICE: 4, 3, 2, 1

Letter	Name	Notes
A	NDREW	company owner speaks Japnies
P	ARVEEN	Translate, Encrypt
S	EEMA	appointment
T	ANVEER	make it down to pieces, packs, sigma
N	NAVEED	two way address, send and receive
D	AVID	seal it
P	EON	physically deliver, send as signal



Section 3

Switching — VLANs, trunking, STP

Day 1

Switch CLI Modes

- User EXEC mode:
  - Show or other basic troubleshooting commands
- Privileged EXEC mode:
  - Full-Access Level
  - Enable Command
- Configuration Mode
  - Config t

Switch configuration using console cable we connect SP of computer the switch console port. Now we enter from our PC – Desktop – Terminal –

```
Switch>
```

This exec Mode if we want to switch to privilege Mode then we write **en**
Or if we want to change Mode to configuration then we write **configure t**

Day 2

Changing Switch Name and password

- Enable en
- Configure terminal confi t
- Hostname KDR
- Enable password *khan*
- exit
- exit
- en
- password *khan*

if we want to remove security or password from switch then we write (**no enable password)** in config mode.

To Show Ports_____

SW> Enable

SW#show ip interface brief **shows all active ports up means connected and down mean not in use**

This is used to shutdown a port _____

Switch#configure terminal

Switch(config)#interface fastEthernet 0/4

Switch(config-if)#shutdown

This is used to shutdown range ports_____

SW#show ip interface brief

Switch#configure terminal

Switch(config)#interface range fastEthernet 0/4 -24

Switch(config-if-range)#shutdown

This is used to up the port_____

Switch#configure terminal

no shutdown

Switch(config)#interface fastEthernet 0/10

6

Switch Port Security Configuration

Switch Port Security Configuration



Switch>enable

Switch#configure terminal

Switch(config)#interface fastEthernet 0/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport port-security

Switch(config-if)#switchport port-security maximum 1

Switch(config-if)#switchport port-security mac-address sticky **we can all so add manually the mac address**

Switch(config-if)#switchport port-security violation shutdown

That some one tray to connect his computer to the secure port that will automatically down

When the secure port is down so that will be no shutdown by administrator.

So when we want to enable that port first we must shutdown then no shutdown

Note: Violation Mode

protect Security violation protect mode
restrict Security violation restrict mode
shutdown Security violation shutdown mode

Note: If we want more then one port so we must to use Range word

Show port-security

```
---------------------------------------------------------------------
        Fa0/1        1          1                    0          Shutdown
---------------------------------------------------------------------
```

Write no shutdown code

Displaying Port security

```
Switch>Enable
Switch#show port-security interface fastEthernet 0/1
------------------------------------------------------------------------------------------------------
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0001.4298.1E4B:1
Security Violation Count   : 0
```

Displaying Port security

```
Switch#show port-security address
                          Secure Mac Address Table
------------------------------------------------------------------------
Vlan  Mac Address       Type                  Ports             Remaining Age
                                                                  (mins)
----  -----------       ------------          ----------        -------------
1     000C.CF25.6890    SecureSticky          FastEthernet0/1     -
------------------------------------------------------------------------------------------------------------------------
Total Addresses in System (excluding one mac per port)     : 0
Max Addresses limit in System (excluding one mac per port) : 1024
==========================================================================
Switch#show port-security
Secure Port MaxSecureAddr  CurrentAddr  SecurityViolation  Security Action
            (Count)        (Count)      (Count)
------------------------------------------------------------------------------------------------------------------------
    Fa0/1     1                1              0                 Shutdown
------------------------------------------------------------------------------------------------------------------------
```

Sticking Mac Address Manually

Ipconfig /all

```
Physical Address.....: 00D0.BAB3.AD86
IP Address...........: 10.0.0.2
Subnet Mask..........: 255.0.0.0
Default Gateway......: 0.0.0.0
DNS Servers..........: 0.0.0.0
```

Copy mac (physical address)

```
Switch>enable
Switch#configure terminal
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum  1
Switch(config-if)#switchport port-security mac-address 00D0.BAB3.AD86        paste mac address here
Switch(config-if)#switchport port-security violation shutdown
exit  exit
```

Set password for Console Cable

```
Switch>
Switch>en
Switch#confi t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#line console 0
Switch(config-line)#password khan
Switch(config-line)#login
Switch(config-line)# exit
```

As long as we connect console switch will ask for the password which is khan

Set password for privilege mode or en mode

```
User Access Verification

Password:

Switch>en
Switch#confi t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#en
Switch(config)#en
Switch(config)#enable se
Switch(config)#enable secret kaka
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
exit
```

Assign IP to switch

```
Config)# interface vlan1
ip add 10.1.1.1   255.0.0.0

no shutdwon
```

```
Switch>en
Password:
Switch#confi t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan ?
  <1-1005>  Vlan interface number
Switch(config)#interface vlan1
Switch(config-if)#ip address 10.0.0.1 255.0.0.0
Switch(config-if)#no shutdown
```

Set vty or telnet password

When someone is trying to enter via cmd

```
Config)#line vty 0 15
Password ccna
Switch#confi t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#line vty 0 ?
  <1-15>  Last Line number
  <cr>
Switch(config)#line vty 0 15
Switch(config-line)#pas
Switch(config-line)#password ccna
```

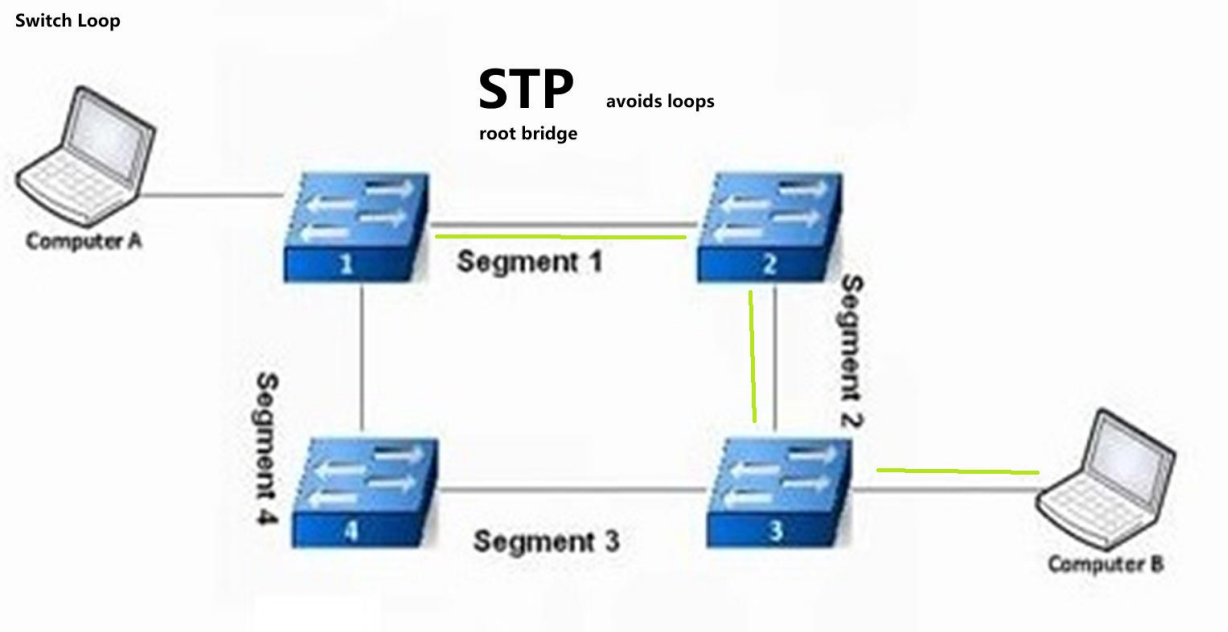
```
PC>telnet 10.0.0.1
Trying 10.0.0.1 ...Open


User Access Verification

Password:
```

Ping to switch via computer cmd will require a password ccna

Switch loop / STP Root Bridge



- Each Switch Has A Bridge ID, (BID) Of Priority Value Followed By MAC Address
- Switches Exchange Bridge Protocol Data Units (BPDUs) To Compare Bridge Ids
- The Switch With The Lowest Bridge ID Becomes The Root Bridge
- Administrator Can Set The Priority To Fix The Selection

STP Commands

Switch>enable

Switch#show spanning-tree

_____ 1

We want made the Root bridge by himself

Switch>enable

Switch#configure terminal

Switch(config)#spanning-tree vlan 1 root primary

Switch#show spanning-tree

_____2

Switch>enable

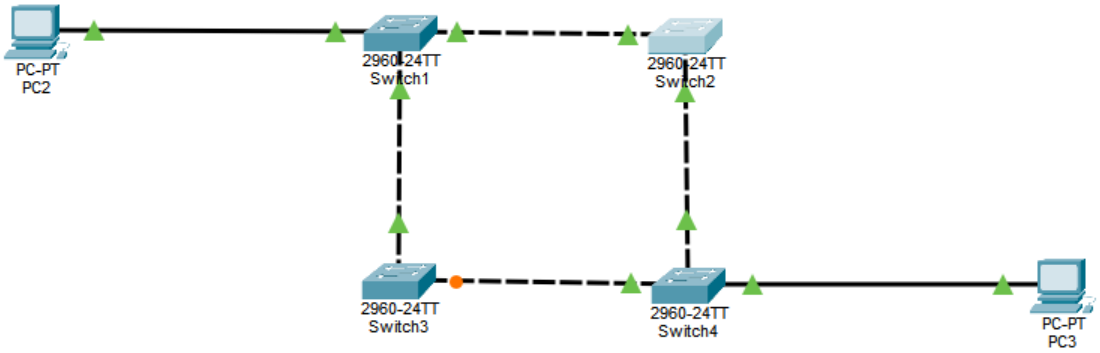
Switch#configure terminal

Switch(config)#spanning-tree vlan 1 priority 20480

Switch#show spanning-tree

RSTP Commands

```
Switch>enable
Switch#configure terminal
Switch(config)#spanning-tree mode rapid-pvst
Switch#show spanning-tree
```



RSTP Commands

```
Switch>enable
Switch#configure terminal
Switch(config)#spanning-tree mode rapid-pvst
Switch#show spanning-tree
```

This is used to select Root Bridge quickly with any delay. So we run this command for all switches.

Save Switch Settings/ Erase

Save the switch's running configuration to NVRAM

```
write memory

copy running-config   startup-config

write erase

reload
banner motd
#########################
@ your Text here @
#########################
```

Work in privilege mode. Used to save switch settings such as name password. Etc. after applying each command we use reload.
Banner motd command works in confi t mode
Banner motd @ this is GT @

VLan

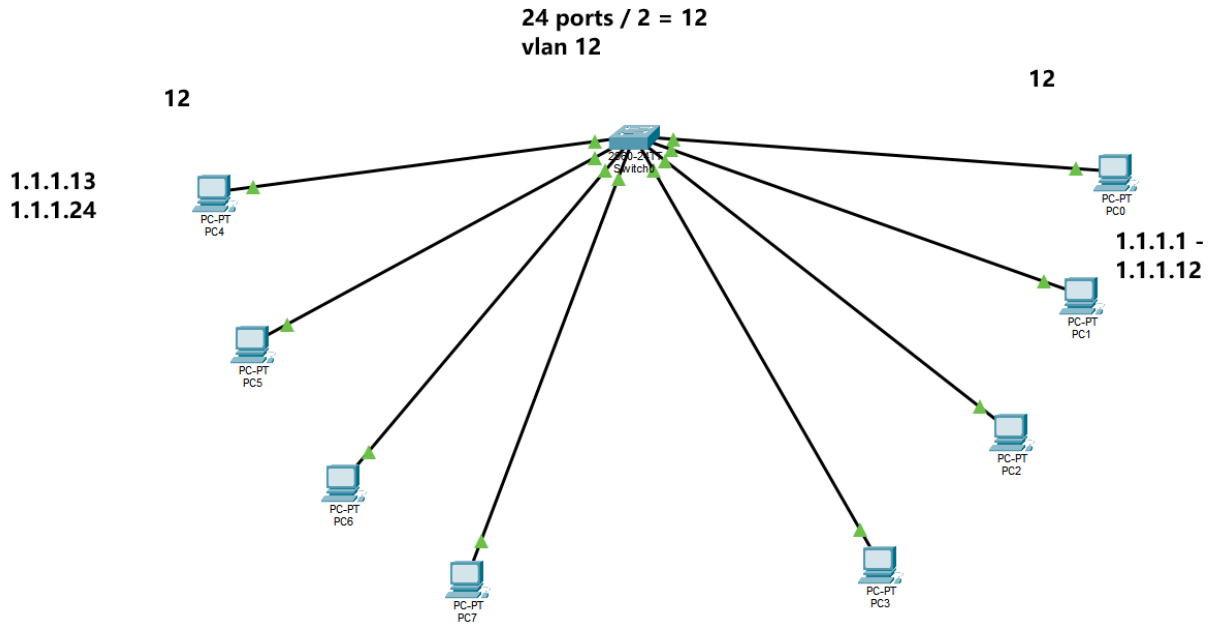
Is used to create different networks in same switch to lower cost.
Cli
Show vlan - show us all available ports
In terminal mode
Vlan 2
Name HR

```
Switch(config)#interface FastEthernet 0/1
Switch(config-if)#SWITchport mode access
Switch(config-if)#switchport access vlan 2
```

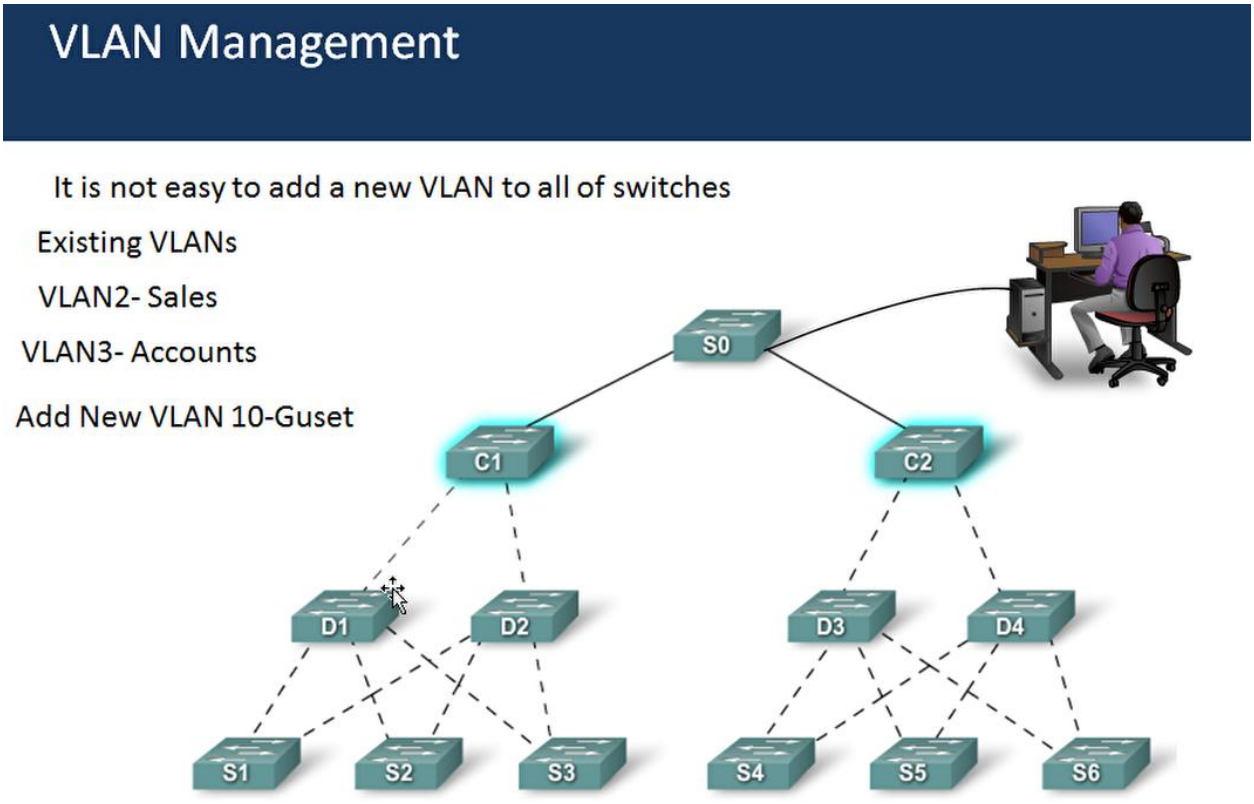
Range

```
Switch(config)#interface range fastEthernet 0/10 -24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 3
```

```
2    HR                                  active    Fa0/1, Fa0/2
3    IT                                  active    Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                   Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                   Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                   Fa0/22, Fa0/23, Fa0/24
```



VTP (Vlan Trunk Protocol)



VTP Server

- One switch acts as server.
- Create VLANs on this switch.
- Information saved in vlan.dat.
- Server sends VLAN information to client switches over active trunk links.
- Add, delete, rename VLANs on server.

Every switch has 3 modes server, client and transparent (only transfer data)

VTP Client

- Client receives VLAN information from server.
- Client switches then have the same VLANs as the server.
- Client does not save VLAN information. It is hold only in RAM and lost if switch is powered off.

We have already 2 VLAN

VLAN 2 Sales

VLAN 3 Accounts

Switch>enable

Switch#configure terminal

Switch(config)#vtp mode server

Switch(config)#vtp domain iqra

Switch(config)#vtp password ccna

Switch(config)#interface fastEthernet0/24

Switch(config-if)#switchport mode trunk

Client Switch

Switch#configure terminal

Switch(config)#vtp mode client

Enter

Switch(config)#vtp domain iqra

Switch(config)#vtp password ccna

Speed

1 Byte = One Char = a or 1

NETWORKING STUDY GUIDE

Bits, Bytes & Internet Speed

b = bit • B = byte • 1 byte = 8 bits

01 Internet speed · Multiples of 1,000

8 bits = 1 Byte = 1 char

Speed	Equivalent rate	Bytes per second
1 kbps	1,000 bits/s	125 B/s
1 Mbps	1,000 kbps	
1 Gbps	1,000 Mbps	
1 Tbps	1,000 Gbps	

02 Data size

Decimal unit	Equals	Binary unit	Equals
1 kilobyte (kB)	1,000 B	1 kibibyte (KiB)	1,024 B
1 megabyte (MB)	1,000 kB	1 mebibyte (MiB)	1,024 KiB
1 gigabyte (GB)	1,000 MB	1 gibibyte (GiB)	1,024 MiB
1 terabyte (TB)	1,000 GB	1 tebibyte (TiB)	1,024 GiB

03 Bits, bytes & characters

Example	Value
1,024 bits	128 bytes
512 bits	64 bytes
Letter A in UTF-8	1 byte. Other characters can take more bytes.
2^7	128
2^{10}	1,024

QUICK CONVERSION

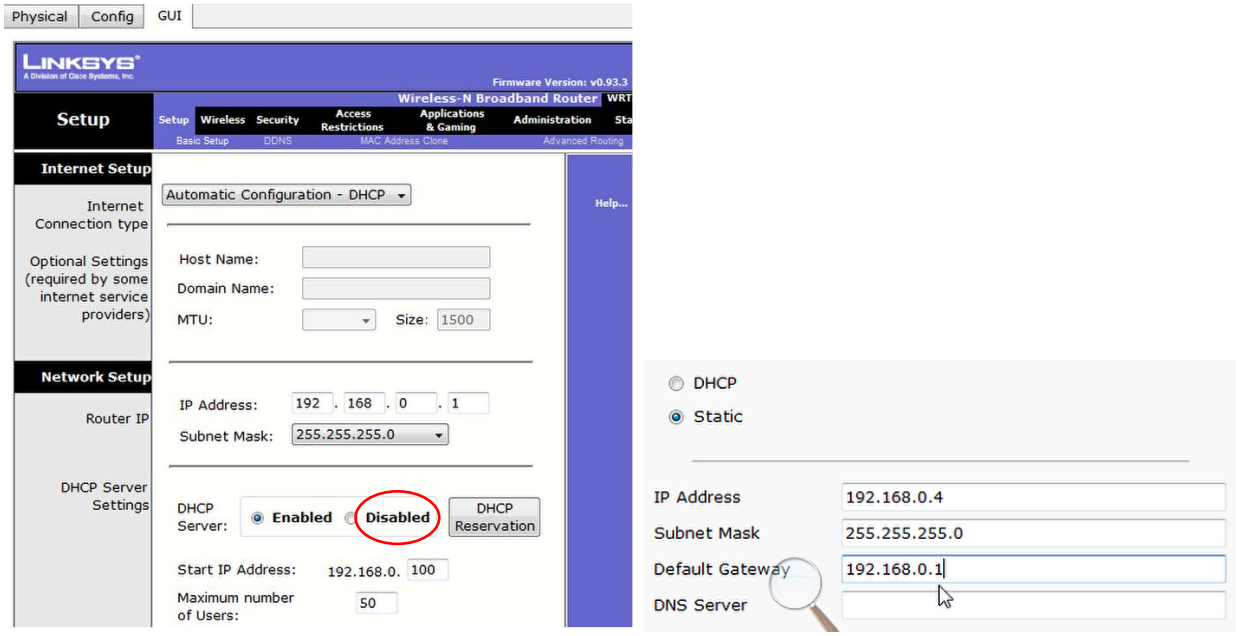
Bits → Bytes ÷ 8 **Bytes → Bits × 8**

Section 4

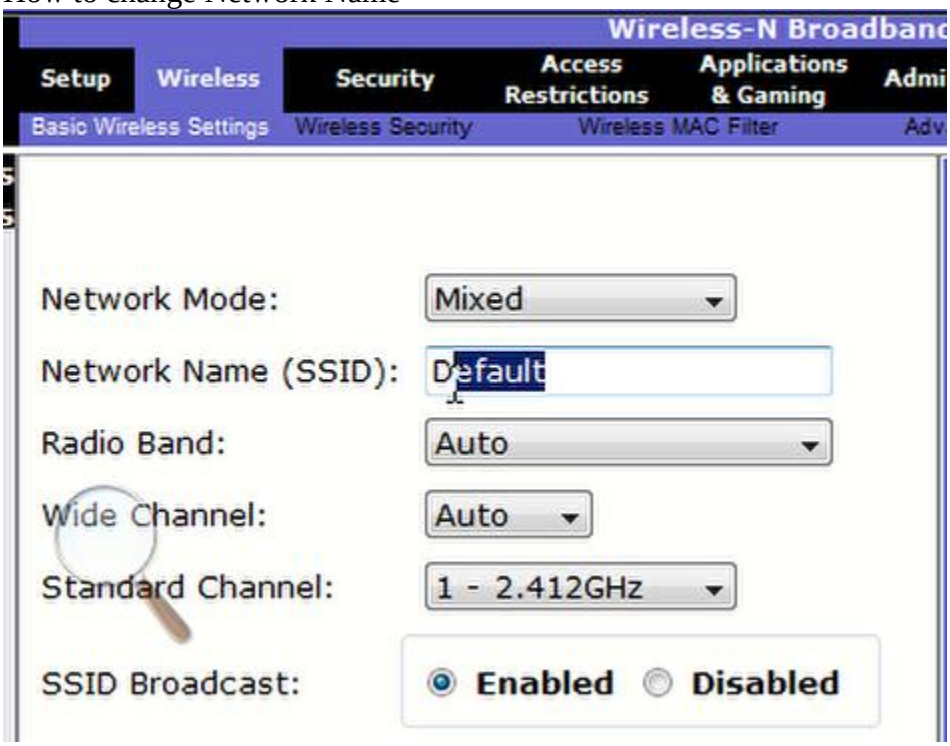
Routing — static routes and OSPF

Router setup with Wireless Network

Routers have **DHCP servers inside which assigns by default IP addresses when it is enabled. And when we make it Disabled then we can assign Static IP means manually entering IPs.**



How to change Network Name



In Administer tab we can change router admin password
In wirless security we can allow or block mac address
How to find mac address cmd – ipconfig/all the physical address is your mac address.

How to configure router from a client browser

Step 1:
Make a wireless connect setup
Step 2:

Click on a laptop/computer/pc you made
Step 3:
Click on desktop then on web browser
Step 4:
Enter your gateway IP
Step 5:
Then ender admin admin
Step 6:
Go to wireless then click wireless security then give it a passphrase
Step 7:
Then connect the computer and you're done
Routing

Routing is how a router uses a packet's destination IP address to choose a path. **Forwarding** is sending the packet along that path.

- **Static route** — you configure it manually.
- **Dynamic route** — the router learns it through a routing protocol such as OSPF.
- **Default route (`0.0.0.0/0`)** — the router uses it when no more specific IPv4 route matches.

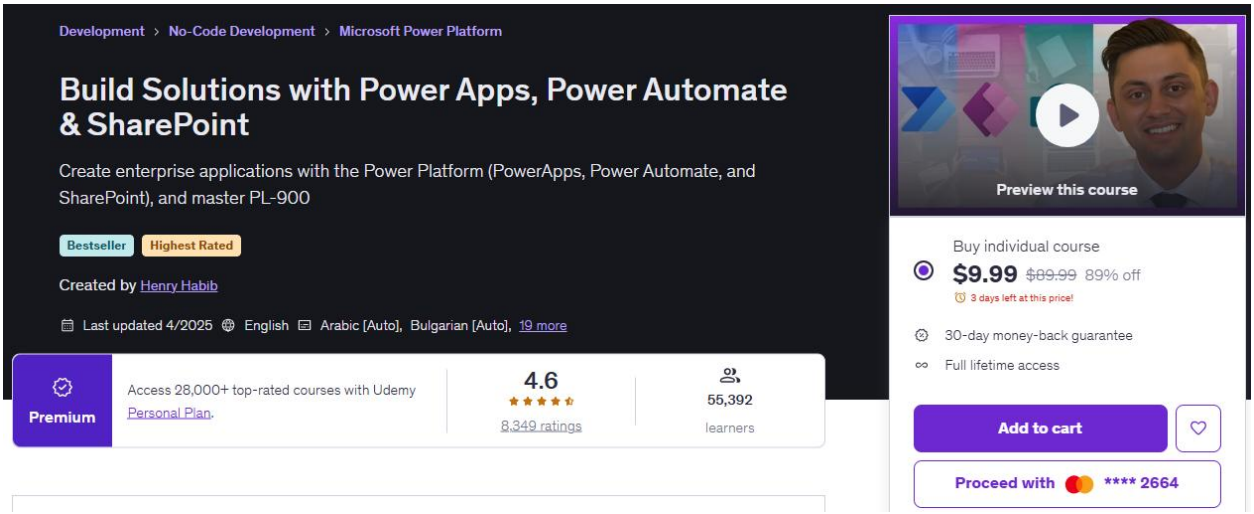
For example, if a packet is headed to **192.168.2.10** and the routing table has a route for **192.168.2.0/24**, the router uses that route to forward the packet.

Wireless networking

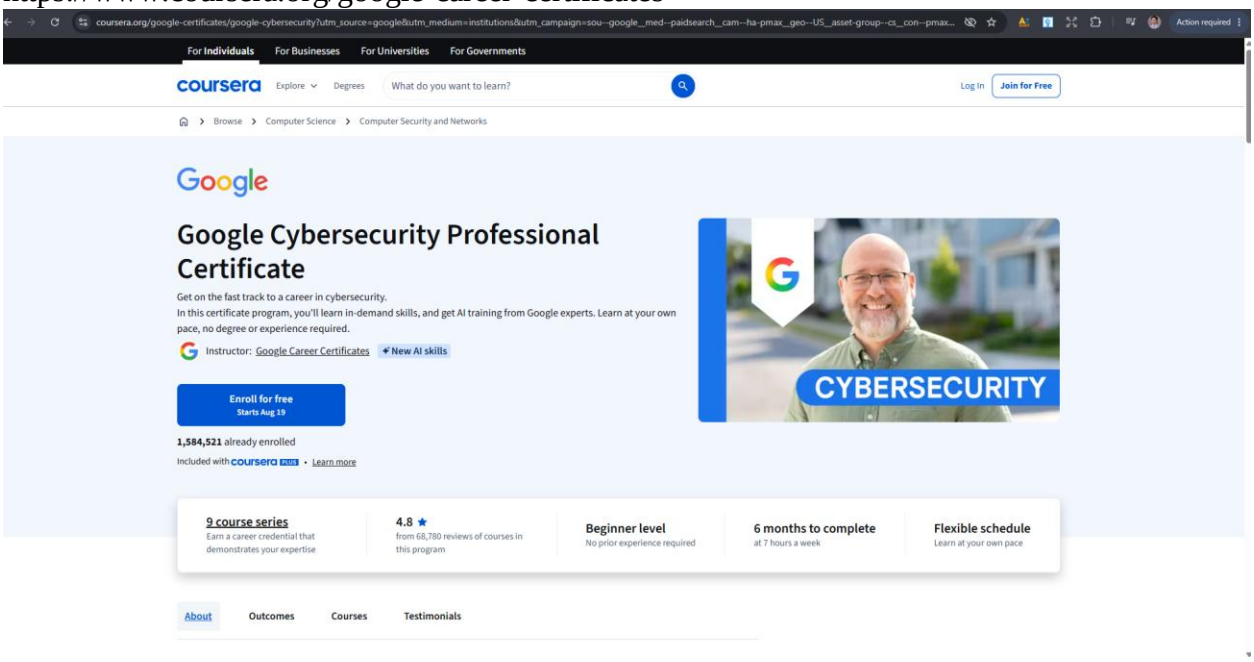
Network security

NAT, DHCP, DNS

Network automation and programmability



https://www.coursera.org/google-career-certificates



https://www.coursera.org/google-certificates/google-cybersecurity?utm_source=google&utm_medium=institutions&utm_campaign=sou--google__med--paidsearch__cam--ha-pmax__geo--US__asset-group--cs__con--pmax__LP--DR__ter--null&gwg_campaign_id=20906443665&gwg_exp=null&gwg_ad_id=GCLID--CjwKCAjwqJXUBhBNEiwA8BgG7vYxmpYotdUV-juPKqYwcXDwTAiRCuJm-j6g_oQjXUZ1ESUE0fEDPBoCGpkQAvD_BwE&_gl=1*1nbigtr*_gcl_aw*R0NMLjE3ODcxNzE5MzUuQ2p3S0NBandxSlhVQmhCTkVpd0E4QmdHN3ZZeG1wWW90ZFVWLWp1UEtxWXdjWER3VEFpUkN1Sm0tajZnX29RalhVWjFFU1VFMGZFRFBCb0NHcGtRQXZEX0J3RQ..*_gcl_au*MjA1NTMyMjQyNC4xNzg3MTcxOTM1